

ORCHESTRA-OS RESEARCH PROGRAM

ORCHESTRA-OS Test Evidence Archive

Retained local test artifacts and integrity record

Evidence boundary: current results are userspace-validated only. This PDF does not establish a kernel scheduler, causal performance advantage, production security, or deployment readiness.

ORCHESTRA-OS test evidence - 2026-08-03

This directory contains every text/data test artifact that remained available locally when the archive was created, plus a fresh full regression capture.

Evidence map

Path	Classification	Interpretation
benchmark/authoritative-c/	Authoritative userspace smoke	Exact copy of <code>/tmp/orchestra-paper-cpu-smoke-v1-20260803-c</code> ; 6/6 validated invocations, source hash <code>d5ec68...8713</code> ; this is the campaign used by the result report and figures.
benchmark/superseded-a/	Superseded	Exact copy of complete six-run campaign <code>-a</code> , source hash <code>b03b78...c55a</code> ; it predates final implementation fixes and must not be mixed with the authoritative result.
benchmark/superseded-b/	Superseded	Exact copy of complete six-run campaign <code>-b</code> , source hash <code>5de386...2193</code> ; it predates final implementation fixes and must not be mixed with the authoritative result.
development-smoke/	Historical development evidence	CSV/log/stdout/stderr files copied from <code>/tmp/orchestra-os-testing.1InLRQ</code> ; provenance is incomplete, so these are debugging records rather than experiment evidence.
regression/final/	Current regression evidence	Fresh <code>make test</code> capture after artifact preservation support was added; includes the combined transcript, machine-readable result, and all five integration scenarios' CSV/log files.
regression/pre-repository-artifact-defaults/	Superseded regression capture	Passing run immediately before the plotting/reporting defaults were changed from <code>/tmp</code> to the retained repository artifact. Retained for completeness.
regression/pre-artifact-index-check/	Superseded regression capture	Passing run with preserved integration CSV/log files immediately before the artifact indexer was added to the top-level syntax checks. Retained for completeness.
regression/pre-integration-artifact-capture/	Superseded regression transcript	Passing run immediately before integration CSV/log preservation was added. Retained so no test result is silently discarded.
regression/pre-capture-tool-check/	Superseded regression transcript	Passing run immediately before the capture tool was added to the top-level syntax checks. Retained for completeness.

Current regression result

`regression/final/test-result.json` records return code `0`, unchanged input hashes, the compiler/tool versions, and the merged transcript hash. The run passed:

- 13/13 named GCC ASan/UBSan unit groups;
- the same 13/13 groups under Clang policy warnings;
- 4/4 benchmark-validator tests; and
- 5/5 integration scenarios covering baseline, ORCHESTRA cadence, tamper rejection, rejected-frame controller gating, and bounded signal teardown.

The regression suite is correctness evidence for the named userspace tests. It does not strengthen the system beyond the **Userspace-validated** claim class.

Copy and hygiene record

- All three benchmark directory copies passed recursive byte comparison against their `/tmp` sources.
- All 22 retained development CSV/log/stdout/stderr files passed byte comparison against their source files.
- Seven generated development test executables were deliberately not copied; they are rebuildable inputs, not test results. Their source hashes are listed below so the exclusion is auditable.
- A text scan found no cookie, authorization, bearer-token, password, API-key, client-secret, access-token, refresh-token, or email-address matches.
- Raw provenance retains the local hostname, username/path, process snapshot, and hardware metadata because this is a local research archive. Review and sanitize a derived copy before public distribution; do not mutate the raw evidence.
- The original `/tmp` directories were not deleted.

Excluded development executables:

File	SHA-256
<code>orchestra_atomic_clang</code>	<code>08ccc1796716e02a3dea367fd866a45209bd856fc75a6d85507bbcb7be2a5ad6</code>
<code>orchestra_atomic_gcc</code>	<code>a051f65088d5f07364967aad6c9936a1cedd2cee347a6853a7f4f4b3fced3c3d</code>
<code>orchestra_release</code>	<code>36520d79649dc240a805fe646bd325d430f2b884eca40479a1ab92897ecd8692</code>
<code>orchestra_san</code>	<code>03e23ad0b6df52f05eaa956622a5fde673efc082a20fd4f0e3a6064c52448f9</code>
<code>orchestra_v2</code>	<code>0aeca6bf63e9a01a6e522c5364306f7aa9f7c72eb8070e89c58c463b8a29884d</code>
<code>orchestra_v2_clang</code>	<code>74fd75d32793fa8b7c097eaa8b8d10eff8cea90473635bf28386c5283d5de7ee</code>
<code>orchestra_v2_strict</code>	<code>3cf28905af15f7455cd8ca8cfa07a004c629a5055690fcbe7e9b1917284f0f01</code>

`SHA256SUMS` covers every retained evidence file except the checksum and inventory files themselves. `inventory.json` records counts, sizes, category totals, and the checksum-manifest identity.